

Test-Time Physical Awareness for Generated Humanoid Reference Motions

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Abstract

Modern humanoid motion trackers such as SONIC can follow rich whole-body reference trajectories in physics, but those trackers still need high-quality target motions. KIMODO makes reference authoring much easier by generating Unitree G1 kinematic motions from text prompts and constraints. This project studies the interface between those systems: given a generated G1 reference, can we detect physical risks, attempt a simple test-time repair, and then gate the result with a controller rollout?

I built a physical-awareness loop for generated reference motions. It evaluates generated references with MuJoCo inverse-dynamics/contact diagnostics [3] and a pretrained SONIC controller audit. The evaluator emits interpretable failure tags such as high torque/root wrench, self-contact, non-foot floor contact, weak support, contact artifact, and controller trackability failure. A deterministic repair baseline creates simple retiming/smoothing variants, and the same physical and SONIC gates decide whether a candidate is accepted, flagged, or rejected.

On the 100-prompt KIMODO benchmark, first-pass references pass the physical screen in 48/100 cases, complete the nominal 4-second SONIC rollout in 53/100 cases, and pass both gates in 29/100 cases. The deterministic repair snapshot improves physical pass from 48/100 to 53/100, SONIC no-fall from 53/100 to 56/100, mean SONIC tracking time from 2.855 s to 3.007 s, and mean RMSE from 0.156 to 0.142. It also has regressions. The contribution is not a solved text-to-robot motion generator; it is a concrete evaluation, repair, and acceptance-gate artifact for exposing generated-reference brittleness.

1 Problem

Humanoid motion control has recently improved because learned policies can be trained to track diverse reference motions under physics. SONIC is a strong example: it trains a Unitree G1 whole-body controller on over 100M frames / 700 hours of high-quality human motion [1]. At the input level, the tracker receives a target trajectory: root pose and joint targets over time.

The upstream question is where those references come from. KIMODO generates controllable kinematic human and robot motions from text prompts and constraints [2]. This makes it a natural source of candidate references for robot training and controller testing. However, a kinematic reference is not automatically physically executable. It may demand unrealistic torques, require impossible support, create self-collisions, drag contacts, or destabilize a learned controller.

The project question is:

Can a physical-awareness layer identify risky KIMODO-generated humanoid reference motions, and can those failure tags guide useful test-time repair before controller execution?

2 System

The final system is a test-time reference screening and repair loop:

Stage	Input	Contribution	Output
Generate	text prompt	KIMODO G1 generation	root + joint trajectory
Evaluate	reference	MuJoCo + SONIC checks	risk flags and rollout metrics
Repair	failed candidate	retiming/smoothing variants	repaired candidates
Repeat/gate	new candidate	rescore until budget ends	accepted or flagged clip

Failure tags are intentionally interpretable. High torque/root wrench signals abrupt motion; self-contact signals limb/body intersection risk; weak support signals a likely balance problem; and trackability failure means that the pretrained SONIC controller could not follow the reference for the rollout horizon. SONIC is used as a foundation-style tracking evaluator and acceptance gate, not as an optimizer that changes KIMODO.

3 Benchmark and Metrics

The benchmark contains 100 humanoid robotics prompts grouped as locomotion and recovery (26), manipulation and loco-manipulation (26), floor/low posture (12), dance/expressive motion (12), athletic and terrain stress (16), and communication/safety gestures (8). The suite includes walking, shuffling, crawling, carrying, panel pressing, gestures, obstacle-style steps, forward rolls, cartwheel attempts, jumping, and low-posture motions.

For each generated G1 trajectory, the evaluator computes torque demand relative to actuator limits, root force/torque, velocity, acceleration, jerk, self-contact, non-foot floor contact, contact artifacts, support-region proxy violations, SONIC no-fall, survival time, and joint RMSE.

4 Results

4.1 First-Pass Audit

Table 1: First-pass 100-prompt KIMODO audit.

Set	Physical pass	SONIC no-fall	Both	Mean sec.	RMSE
All KIMODO refs	48/100	53/100	29/100	2.855	0.156
Physical-pass subset	48/48	29/48	29/48	3.293	0.170
Flagged subset	0/52	24/52	0/52	2.450	0.143

Only 29 of 48 physical-pass references also complete SONIC. Meanwhile, 24 of 52 flagged references still complete SONIC. This disagreement is a central result: reference-level physical checks and controller-level trackability are both needed.

4.2 Failure Tags

Table 2: First-pass non-exclusive failure tags.

Failure tag	Count
Physical screen fail	52
SONIC fall	47
Torque limit >1x	66
High root force >5 kN	28
Self-contact >8%	34
Non-foot floor contact	11
Low foot support	7
Contact artifact >0.45	15
Floor penetration >8 cm	10

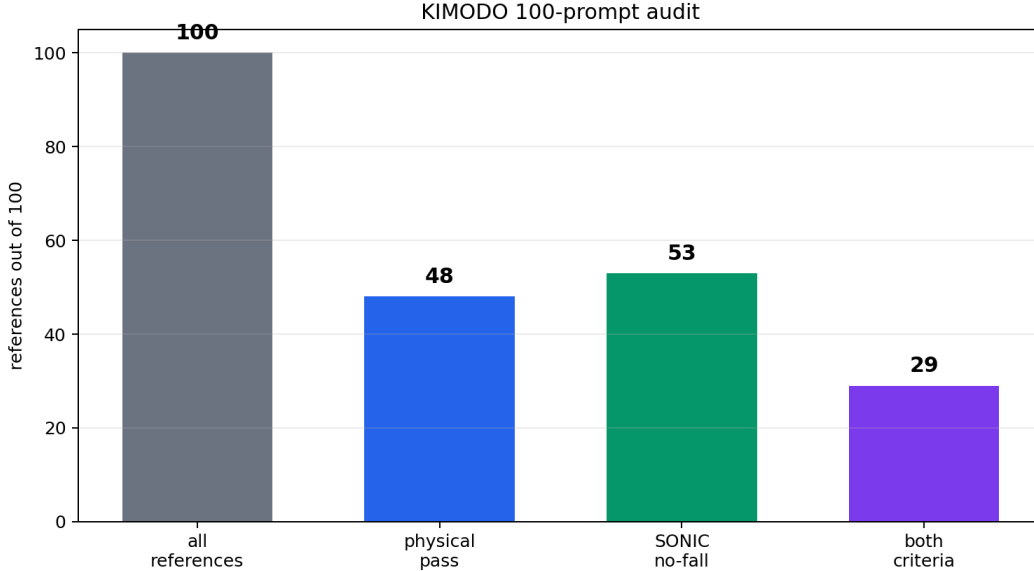


Figure 1: First-pass KIMODO audit over 100 prompts. The physical screen and SONIC controller rollout are related but not equivalent.

The representative failure videos in the deck are ‘forward_roll’ for torque/root wrench, ‘turn_in_place’ for self-contact, ‘knee_slide’ for non-foot floor contact, ‘step_over_right’ for support proxy, ‘carry_box’ for controller trackability, and ‘happy_dance’ for contact artifact.

4.3 Deterministic Repair Snapshot

Table 3: Measured retiming/smoothing repair snapshot.

Metric	Original	Repaired
Physical pass	48/100	53/100
Critic accept	47/100	54/100
Reject/regenerate	18/100	16/100
SONIC 4 s no-fall	53/100	56/100
Mean SONIC seconds	2.855	3.007
Mean RMSE	0.156	0.142
Mean risk	41.548	37.916
Contact artifact	0.264	0.247

The repair pass produced seven SONIC rescues and four regressions. The main repair videos show representative improvements: ‘forward_walk’ becomes slower and smoother after a trackability flag; ‘wave’ uses planted feet and smooth arm motion after torque/tracking flags; ‘split_squat_jump’ uses a smaller jump and soft landing after support/torque flags; ‘happy_dance’ improves contact placement; and ‘backward_recovery’ uses a smaller lean and step back. The deck also includes an illustrative ‘pushup_pose’ height-clearance demo; the table above is the measured evidence.

5 Interpretation

The results support the hypothesis in a bounded way. Generated references have measurable physical failure modes; failure tags are useful for curation and simple repair; and SONIC is stricter than kinematic replay. Human visual review is used as a sanity check for presentation examples, not as the source of metric labels.

The boundary slide shows remaining failures: ‘forward_roll’ for torque/root wrench stress, ‘carry_box’ for controller trackability, and ‘cartwheel_attempt’ for acrobatics. The current method does not solve arbitrary text-to-robot motion, robust acrobatics/low-posture tracking, or automatic repair for every invalid reference. Stronger results likely require fine-tuning or training a motion-level refiner, not only retiming and smoothing.

6 Limitations and Future Work

Deterministic retiming/smoothing repair helps some clips but is not a general motion generator. The physical screen is heuristic and simulator-dependent. SONIC rollout is a simulation diagnostic, not real hardware validation. Some motions need semantic context: floor contact can be correct for crawling and wrong for walking. Prompt fidelity is checked qualitatively; there is no full human-study or vision-language benchmark.

Future work should train a risk-conditioned refiner that learns from failure tags and SONIC rollouts, use short controller rollouts or a learned surrogate during candidate selection, and improve semantic evaluation so physical stability does not hide prompt mismatch.

7 Team Responsibilities

Tae Hoon Yang completed the project solo: prompt suite design, KIMODO/SONIC integration, MuJoCo physical checks, repair loop, video rendering, slide deck, and final report.

References

- [1] Zhengyi Luo, Ye Yuan, Tingwu Wang, Chenran Li, Sirui Chen, Fernando Castañeda, Zi-Ang Cao, Jiefeng Li, David Minor, Qingwei Ben, Xingye Da, Runyu Ding, Cyrus Hogg, Lina Song, Edy Lim, Eugene Jeong, Tairan He, Haoru Xue, Wenli Xiao, Zi Wang, Simon Yuen, Jan Kautz, Yan Chang, Umar Iqbal, Linxi Fan, and Yuke Zhu. Sonic: Supersizing motion tracking for natural humanoid whole-body control, 2025.
- [2] Davis Rempe, Mathis Petrovich, Ye Yuan, Haotian Zhang, Xue Bin Peng, Yifeng Jiang, Tingwu Wang, Umar Iqbal, David Minor, Michael de Ruyter, Jiefeng Li, Chen Tessler, Edy Lim, Eugene Jeong, Sam Wu, Ehsan Hassani, Michael Huang, Jin-Bey Yu, Chaeyeon Chung, Lina Song, Olivier Dionne, Jan Kautz, Simon Yuen, and Sanja Fidler. Kimodo: Scaling controllable human motion generation, 2026.
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